SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Scenario Based Assignment — Answers

Code of Conduct:

I ___Prabhavathi R (192321101)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Annexure A — Scenario Based Assignment

1. Emergency Medicine Delivery Using Search Algorithms

i. Greedy Best-First Search

Greedy Best-First Search selects the route with the smallest heuristic value, usually the straight-line distance to the destination. It chooses the path that appears closest to the delivery point.

Strength: It can find a route quickly. Weakness: It ignores the cost already travelled and may select a blocked, risky, or weather-affected route. It is not guaranteed to find the shortest or safest path.

ii. A* Search

A* uses both the cost already travelled and the estimated remaining distance. Its evaluation function is:

$$f(n) = g(n) + h(n)$$

Here, g(n) is the actual travel cost so far and h(n) is the estimated cost to the goal. A* balances route cost and estimated distance and can find a lower-cost route when its assumptions are suitable.

iii. Impact of heuristic accuracy

An accurate heuristic guides search toward the goal. Greedy search depends heavily on h(n), so a misleading estimate may lead it in the wrong direction. A* also considers g(n). An overestimated heuristic can affect A* optimality.

iv. Effect of dynamic changes

If a path becomes blocked, the current route may no longer be valid. The drone must use updated information and re-plan. Frequent changes can increase search time.

v. Recommendation

A* with frequent updates and re-planning is suitable for finding a low-cost route. For urgent deliveries, use a time-limited A* variant and include risk and weather in the path cost. Safety checks and current obstacle information are essential.

2. Smart City Traffic Signal Optimization

Formulation: Treat each signal timing plan as a candidate solution. Minimize a combined cost of waiting time, fuel consumption and congestion, subject to road capacity and signal safety constraints. Exhaustive search is inefficient because the number of timing plans grows rapidly and traffic changes continuously. **i. Hill Climbing**

Hill Climbing starts with a timing plan and repeatedly selects a neighbouring plan that improves the objective. It is simple and fast, but may stop at a solution that is locally good rather than globally best.

ii. Local maxima, plateaus and ridges

Local maximum: a plan looks best among nearby options but a better plan exists elsewhere. Plateau: many plans have nearly equal cost, so progress stops. Ridge: improvement requires coordinated changes that one-step changes do not show. For example, changing one traffic signal may not help unless nearby signals are adjusted too.

iii. Simulated Annealing or Genetic Algorithms

Simulated Annealing sometimes accepts a worse move early in the search, helping it escape local optima; this becomes less likely as temperature decreases. A Genetic Algorithm maintains a population of plans and uses selection, crossover and mutation to explore combinations.

iv. Recommendation

A Genetic Algorithm is suitable for large networks because it explores many timing plans and can optimize multiple objectives. Use current traffic data and periodic re-optimization. Simulated Annealing is an alternative when computing resources are limited.

3. Autonomous Rover Exploring Mars

i. Why online search is needed

The rover does not have a complete map before starting. It must act, observe the terrain and choose its next action. Online search discovers information while moving and updates the plan as new information arrives.

ii. Partial observability and dynamic environment

Sensors see only part of the surroundings and may be inaccurate. Hidden craters, rocks, slopes and communication delays create uncertainty. The rover must avoid unsafe actions and re-plan when observations change.

iii. Building and updating an internal model

The rover combines sensor readings, camera images, movement data and previous observations to build a local map. It marks safe areas, obstacles and unexplored regions, then updates the map after each movement.

iv. Online, uninformed and informed search

Uninformed search has no heuristic guidance and may explore many unnecessary states. Informed search uses a heuristic, such as estimated distance to a target. Online search emphasizes learning and acting in an unknown environment; it can also use informed heuristics.

v. Recommendation

Use an online, heuristic-guided agent with frequent sensing, local mapping, obstacle avoidance and re-planning.
It adapts to unknown terrain and supports safer decisions, even if the safest route is longer.

4. University Exam Timetable Using CSP

A Constraint Satisfaction Problem (CSP) consists of variables, domains and constraints.

i. Variables, domains and constraints

Variables: each course's exam time slot and hall. Domains: available time slots and halls. Constraints: a student cannot have overlapping exams; hall capacity must be sufficient; required subject order must be followed; faculty must be available; and special accommodations must be satisfied.

ii. Backtracking search

Backtracking assigns a time slot and hall to one exam at a time. If an assignment violates a constraint, the system goes back and tries another value. It continues until a valid timetable is found or all choices are exhausted.

iii. Constraint propagation and forward checking

After assigning a slot to an exam, forward checking removes incompatible slots from the domains of related exams. This detects conflicts early and reduces unnecessary searching.

iv. Challenges and scalable strategy

The search grows as courses, students and constraints increase. Use Minimum Remaining Values (MRV) to select the exam with the fewest legal choices, degree heuristic to break ties, and least-constraining value to choose a slot. Combine these with forward checking for better scalability.

5. Strategic Game AI: Minimax and Alpha-Beta Pruning

i. How Minimax models the game

Minimax represents possible moves as a game tree. The AI maximizes the score, while the opponent minimizes it. It assumes both players choose rationally.

ii. Computational challenges

If each state has b possible moves and the search depth is d, the number of nodes can grow approximately as O(b^d). Large branching factors and deep searches make real-time decisions difficult.

iii. Alpha-Beta pruning

Alpha-Beta pruning skips branches that cannot change the final decision. Alpha is the best value found for the maximizing player; beta is the best value for the minimizing player. When alpha is greater than or equal to beta, the remaining branch can be pruned. It returns the same decision as Minimax at the searched depth while evaluating fewer nodes.

With good move ordering, the best-case time complexity approaches O(b^(d/2)); the worst case remains O(b^d). **iv. Evaluation function**

A simple evaluation function is:

$$\textbf{Score} = w1(\text{Material}) + w2(\text{Position}) + w3(\text{Objectives}) - w4(\text{Damage}) - w5(\text{Resource Cost})$$

Material measures remaining units, Position measures control of useful locations, Objectives measures progress, Damage measures losses, and Resource Cost measures resources spent. The weights represent each factor's importance.

v. Recommendation

Use iterative deepening Minimax with Alpha-Beta pruning, good move ordering and a heuristic evaluation function. Set a strict time limit and return the best move found so far when time runs out. This balances decision quality with real-time performance.